

Real-time detection tracking and recognition algorithm based on multi-target faces

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Abstract

At present, face recognition algorithms are facing some problems with poor face tracking and low real-time performance in multi-target recognition scenarios. This paper details a multi-target face real-time detection tracking and recognition algorithm, including three methods of fast-tracking, fast detection, and quick recognition. The first step offers a new network based on GOTURN for achieving fast face tracking. The prior information of the previous frame image used to predict the position of the face boxes at the current frame. The second step is based on MTCNN for face detection, using the prior information of the present structure to avoid generating massive of invalid candidate boxes, thereby achieving rapid detection of faces. Finally, fast face recognition realized by reduced MobileFaceNet. By avoiding repeated exposure and repeated identification of the same target, the algorithm successfully transforms a multi-target scene into a single-target scene. On the OTB2015 and 300_VW test sets, the evaluation trackers tracked faces with an accuracy rate of 92.2% and 99.6% respectively. On the Xiph test set, multi-target detection and tracking face speed reached 102fps on the CPU. Compared with the original MobileFaceNet, the streamlined network has an accuracy rate of 99.1% on LFW, the feature extraction speed increased by 25%, and the model size reduced by 45%. Experimental results show that the algorithm has high recognition accuracy and real-time performance in multi-target recognition scenes.

Keywords Low real-time · Fast detection · Fast tracking · Fast recognition · Multi-target face

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1 Introduction

In the past few years, deep learning made a great progress in the field of image recognition due to the rapid development of artificial intelligence. The accuracy and tracking speed of face detection has been improving continuously, which satisfied the use in single target recognition scenarios. However, in the multi-target recognition scene, the increment of the face amount causes a considerable increase in the amount of calculation, which easily leads to a speed explosion. At present, the target tracking algorithms based on deep learning gains high accuracy while the tracking speed is too slow to achieve real-time effects. Therefore, the rate of detection and monitoring of multi-target faces still need to be accelerated.

Compared with traditional face recognition algorithms, deep learning-based face recognition algorithms achieves better performance in image recognition tasks. Convolutional Neural Networks [20] can learn high-dimensional semantic features and in-depth features from a massive amount of samples. CNN (Convolutional Neural Networks) is a pioneering work in computer vision. After CNN was proposed, many extension methods appeared continuously for improving the speed and stability of image detection algorithms. In 2014, Yaniv Taigman et al. proposed DeepFace,[23] which uses CNN to extract in-depth face similarities, improving the accuracy of recognition. In 2015, Kaiming He and his colleagues proposed Residual Neural Networks,[10] which solved the problem that the network gradient disappeared after the CNN network reached a certain depth. In 2016, Kaipeng Zhang and others proposed the MTCNN (Multi-Task Cascaded Convolutional Neural Networks)[28] face detection algorithm, which uses three cascaded CNN networks to detect face positions and critical points, with excellent results and fast speed. In 2017, Weiyang Liu and others proposed the SphereFace[18] algorithm, which improved the loss function so that the algorithm can learn more separable angular characteristics. In 2018, Jiankang Deng and others proposed the ArcFace[5] algorithm. The ArcFace algorithm paid more attention to distinguishing categories in angular space. After those improvements, the accuracy of face recognition has reached a very high level.

Object detection and tracking in the video are essential research topics. Compared with static images, videos contain more abundant information. However, the recognition tasks in videos are complicated and require high real-time performance, so it has higher requirements on algorithms and equipment. Prior to 2010, the most commonly used methods in the field of target tracking were traditional tracking methods, such as Mean Shift, particle filtering, Kalman filtering, and feature-based optical flow algorithms.[16] these methods were not able to adapt to complex motion changes in the video. In 2012, Henriques et al.[12] They have proposed a cyclic structure kernel tracking method based on a cyclic matrix and a kernel function. The cyclic matrix is used to complete dense sampling, and the Fourier transform is used to achieve fast detection. Henrique and others subsequently proposed an improved CSK algorithm, kernel correlation filtering / discrimination correlation filtering, which improved the tracking accuracy while ensuring high-speed processing. In 2013 and 2015, Wu Y et al.[25, 26] proposed that OTB50 and OTB100 (Object Tracking Benchmark) provide standards for unified evaluation trackers, which directly promoted the development of tracking algorithms. The Hyeonseob et al.[19] multi-domain network (MDNet) method in 2015 used multiple types of tracking sequences to pre-train the

network and fine-tune the model during online tracking. In 2016, GOTURN (Generic Object Tracking Using Regression Networks) proposed by David Held et al.[11] Which achieved a speed of 100fps on the GPU with the poor accuracy. In comparison, related filtering algorithms have faster-tracking rate, and deep learning methods have higher accuracy.

This paper proposes a multi-target face real-time detection tracking and recognition algorithm based on deep learning, which maintains high tracking accuracy and accelerates the tracking speed. The algorithm includes three strategies: fast-tracking, fast detection, and quick recognition. This paper designs a new network structure, FTNet (Fast track Net), which is based on GOTURN, But FTNet has a simple structure, high tracking accuracy and fast-tracking speed, First, face detection is performed based on the MTCNN algorithm, and the information of the detected initial face frame is used to obtain the face boxes information of the current structure by using FTNet. The distance between the center of the new and old face boxes is used to determine whether it is a new target, It can help to avoid repeated recognition of the same object. At the same time, set all the pixel values of the detected face area in the picture to 0 by using the position information of the face boxes. It can help to avoid generating a lot of invalid candidate frames, In a multi-target recognition scenario, the more amount of targets, the more sparse the matrix. It can help to save the time for filtering candidate boxes and speed up the overall face detection speed. After a successful face detection, The features of face will be quickly extracted by using the reduced MobileFaceNet[2] .And MMFNet (Min MoblieFaceNet) trained by the Arcface loss function. The algorithm only needs to detect and recognize each face once, and only needs to track the detected faces through FTNet, which successfully transforms a multi-target scene into a single-target scene. Finally, the accuracy rates of FTNet through the test set of the OTB2015 and 300-VW test sets were 92.2% and 99.6%, respectively. The FTNet tracking and fast detection speed through the Xiph's multi-face test set reached 102FPS on the CPU, and MMFNet was evaluated on the LFW. The accuracy rate is 99.1%, the volume is reduced by 45%, and the speed is increased by 25%. Experiments show that the multi-target real-time detection tracking and recognition algorithm proposed by this paper also has high accuracy and real-time performance in multi-target recognition scenarios.

2 Related work

2.1 Face recognition system

Referring to Fig. 1, the face recognition system consists of two parts. The first part is face recognition which includes face detection, face feature extraction and feature matching. The main purpose of the second part is target tracking on the recognized faces. Due to delay caused by the first part face recognition, especially in a multi-target environment, performing face recognition on each frame of video will cause the system to freeze or crash. After face recognition, it usually converted to target tracking for face. And the feature check is performed to check the target face to ensure the accuracy of tracking at the last section.

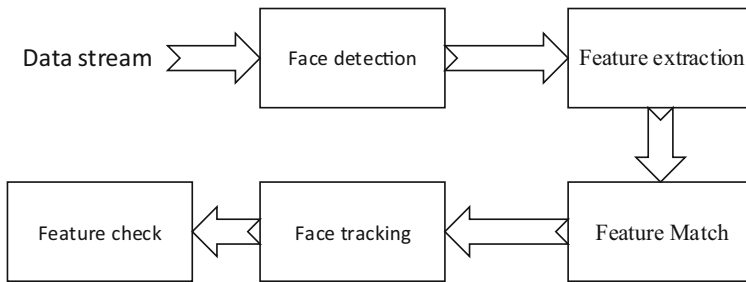


Fig. 1 Face recognition system diagram

2.2 Face recognition

2.2.1 Face detection

MTCNN is an algorithm that uses multi-task cascaded convolutional neural systems for face detection. The algorithm uses three cascaded networks, PNet (Proposal Network), RNet (Refine Network), and ONet (Output Network). By reducing the number of filters, setting a small convolution kernel and increasing the depth of the network structure, higher-level semantic information can be obtained in face detection. Compared with traditional manual feature-based detection methods, the algorithm has better robustness.

The algorithm first transforms the images at various scales to build an image pyramid to detect faces of different sizes. Then, several candidate frames generated on the image at different levels, and the probability that all candidate frames are faces is predicted by the PNet network. Using NMS(Non-Maximum Suppression) in all candidate boxes for filtering out candidate boxes with higher probability. Through the RNet network, the position and size of candidate boxes screened by PNet are regression-corrected. And the probabilities of the candidate boxes are predicted by PNet. In the same way, using the NMS processing in all candidate boxes to filter out the boxes; finally, the ONet is employed to further regression correct and NMS process the position and size of the prediction frame screened by the RNet network. At the same time, each prediction boxes can obtain the coordinate information of five points of two eyes, the tip of the nose, and the corners of the mouth on both sides.

2.2.2 Feature extraction

Literature [7] proposes a reduced network structure MMFNet, which based on MobileFaceNet. And MobileFaceNet is a lightweight CNN network with high accuracy for extracting features. Depthwise Separable Convolution[3] is employed to reduce network parameters, significantly reducing the amount of calculation, making the network more suitable for mobile devices. Meanwhile, MobileFaceNet uses MobileNetV2[21] to replace the Global Average Pooling layer with a separable convolution layer, so that the network can learn the weights of different points. The structure of MMFNet shown in Table 1 is different from the feature vectors of the last 512 feature points of MobileFaceNet. MMFNet finally uses 128 feature points as the feature vectors. The channels Shuffle strategy[29] is added to MMFNet. It can

Table 1 MMFNet Architecture

Input	Operator	MMFNet
$112 \times 112 \times 3$	data	
$56 \times 56 \times 64$	conv1	3×3 , 60, stride = 2
$56 \times 56 \times 64$	conv2_dw	deepwise, 3×3 , 60
$28 \times 28 \times 64$	conv3_x	conv, 1×1 , 120 deepwise, 3×3 , 120 channel shuffle, g3 conv, 1×1 , 60} $\times 3$
$14 \times 14 \times 128$	conv4_x	conv, 1×1 , 240 deepwise, 3×3 , 240 channel shuffle, g3 conv, 1×1 , 120} $\times 3$
$7 \times 7 \times 128$	conv5_x	conv, 1×1 , 240 deepwise, 3×3 , 240 channel shuffle, g3 conv, 1×1 , 120} $\times 3$
$7 \times 7 \times 128$	conv6_sp	conv, 1×1 , 480
$1 \times 1 \times 128$	fc1	deepwise, 7×7 , 128

strengthen the information flow between channels and enhance the information expression ability of the network.

The loss function ArcFace chosen by MMFNet maximizing the classification boundary in angular space. The idea of ArcFace and AMSoftmax have certain common points, the difference is: ArcFace maximizes the classification limit in angular space, and AMSoftmax maximizes the classification limit in cosine space. The ArcFace loss function is obtained based on the modification of the traditional Softmax. The Softmax shown in formula (1), and the loss function of the ArcFace algorithm shown in formula (2).

$$L_1 = -\frac{1}{m} \sum_{i=1}^m \log \frac{e^{W_{yi}^T x_i + b_{y_i}}}{\sum_{j=1}^n e^{W_{ji}^T x_i + b_j}} \quad (1)$$

$$L_3 = -\frac{1}{m} \sum_{i=1}^m \log \frac{e^{s \cdot \cos(\theta_{y_i} + m)}}{e^{s \cdot \cos(\theta_{y_i} + m)} + \sum_{j=1, j \neq y_i}^n e^{s \cdot \cos \theta_j}} \quad (2)$$

2.2.3 Feature matching

After the face features are extracted, the features need to be matched in the feature database to find the corresponding identity information. A common method is to calculate the similarity between the target face feature and all the features in the feature library, followed by picking the most consistent feature according to the similarity. In this paper, cosine similarity is used to represent the similarity of features. X and y are the feature vector of the two face. $\cos \theta = \frac{x \cdot y}{\|x\| \cdot \|y\|}$. The greater the $\cos \theta$, the higher the similarity of the two face.

2.3 Target tracking

GOTURN is the first generic object neural-network tracker that is able to run at 100 fps. GOTURN only passes two images through the network, and the network regresses directly to the bounding box location of the target object.

The target of the previous frame in the video and the search Region of the current frame are sent to the convolutional layer simultaneously, and then passes the output of the convolutional layer to the fully connected layer, which is used to return to the position of the target of the

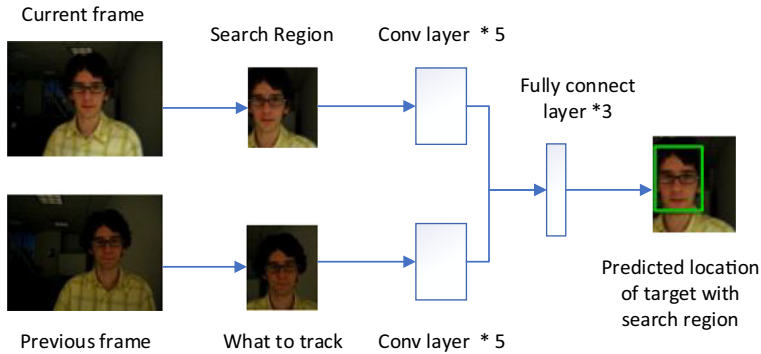


Fig. 2 The Architecture of GOTURN

current frame. The implementation process is shown in the Fig. 2. The entire framework can be divided into two parts: the convolutional layer is used to extract the features of the target region and the search region, and the fully connected layer is used to compares the characteristics of the target region and the search region, and then acquire a new target position.

The convolutional layer of the network uses the first 5 layers of caffeNet, followed by 3 fully connected layers. Each layer has 4096 nodes. After the fully connected layer is an output layer with only 4 nodes, it is used to output the coordinates of the upper left and lower right of the target.

3 Real-time detection tracking and recognition algorithm based on multi-target faces

This paper first uses MTCNN algorithm to detect faces, and then uses the affine transformation in the face boxes filtered by MTCNN to achieve face alignment, and extracts features from the aligned faces through MMFNet network after the extraction, the cosine similarity is used to match elements in the feature database. Finally, the target is tracked by FTNet.

MTCNN is a cascade-based specific target detector that widely used at present, and it is also one of the few detection algorithms that can land on traditional hardware. In the face recognition ~~work in the~~ industry, MTCNN points are often used for alignment. In the task of face detection, the accuracy and precision of face recognition are relatively high when the face base are less than 4, but as the number of the face increases, it often leads to exploding in speed. The experiment shows that the bottleneck of MTCNN speed mainly divided into two points. The first multi-scale face candidate frame selection takes a long time with generating an image pyramid in the PNet network. The second is that the calculation time of ONet and RNet networks will increase due to the increase in the number of faces. In response to these problems, this paper proposes a fast detection strategy and a fast target tracking algorithm.

3.1 Fast face tracking algorithm

In current face recognition systems, there are repeated detections and repeated recognitions of the same target. Through the target tracking algorithm, repeated recognitions of the same target can be solved. The improved target tracking algorithm based on MTCNN determines whether

the target is the same by comparing the positional relationship of candidate frames in the two adjacent pictures. But it is unavoidable to detect the same target repeatedly, and when the number of the face is too large, the amount of calculation increases significantly, resulting in a reduction in speed. The FTNet proposed in this paper directly uses the position information of the face boxes in the previous frame as the initial position to correct the position information of the face boxes in the current frame. By comparing the correction distances of the face boxes twice, it is determined whether they are the same face. Once the target is detected successfully, it only needs to be tracked through FTNet. This strategy also avoids repeated detection and repeated recognition of the same target. It improved the real-time performance of the algorithm significantly. The FTNet structure is shown in Fig. 3. The first two bits of the output indicates whether the target boxes have lost, and the last four bits indicate the correction amount of the candidate boxes. FTNet is a shallow network that cascades two FTNet and then uses the position information of the face boxes of the previous image frame to correct the face boxes of the current structure. The calculation speed is fast, and the accuracy is high.

FTNet is designed based on the idea of GOTURN. The input of FTNet is the search region in the current frame. The search region is determined by the target position of the previous frame. The input of GOTURN is the target of the previous frame and the search region in the current frame. Compared with GRTURN, the input of FTNet is simpler and the network depth of FTNet is shallower. Therefore, the speed of tracking is greatly improved.

FTNet's loss function consists of Box lost loss and Box regression loss. Box lost loss is shown in Eq. (3). The learning objective is formulated as a two-class classification problem. For each sample, we use the cross-entropy loss: y_i^{det} denotes the ground-truth label., $y_i^{\text{det}} \in \{0, 1\}$, p_i is the probability that the face is not lost.

$$L = -(y_i^{\text{det}} \log(p_i) + (1 - y_i^{\text{det}})(1 - \log(p_i))) \quad (3)$$

Box regression loss is shown in Eq. 4. For each candidate window, we predict the offset between it and the nearest ground truth (i.e., the bounding boxes' left, top, height, and width). The learning objective is formulated as a regression problem, and we employ the Euclidean loss for each sample. is obtained through network prediction, y_i is the actual real background coordinates. There are four coordinates, including left top, height and width, and thus y is a quad (upper left corner x , upper left corner y , length, width).

$$L = \|\hat{y}_i - y_i\|_2^2 \quad (4)$$

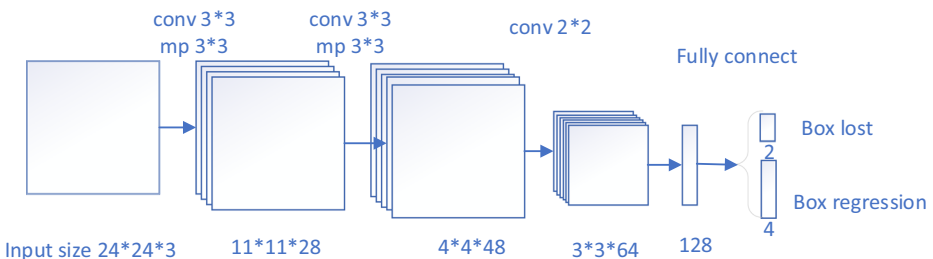


Fig. 3 FTNet Architecture

3.2 Fast detection strategy

When the MTCNN algorithm generates candidate boxes through the PNet network, the feature points of each picture on the image pyramid will create a candidate box. This strategy ignores the face boxes of information that has detected in the previous frame. The fast detection strategy proposed in this paper is to reduce the generation of candidate boxes by using the prior knowledge obtained during the last frame. First, setting the pixel value of the face frame area obtained by the FTNet to 0. It can help to ensure the detected face will not be repeatedly recognized. Then, dividing the detection image into three areas vertically, according to the number of frames of parity, Set the pixels of the first and third frames to 0. This strategy significantly reduces the generation of candidate boxes and the repeated detection of the same target, thereby speeding up the target detection speed.

Figure 4 is the detection process of the first three frames of the video. The complete detected image area is marked as A. After the first frame is detected by MTCNN, A is divided into three areas: B, C, and D. The area of face boxes is marked as E, and using the FTNet to correct the coordinates of the face boxes of the first frame for obtaining the new coordinates of the face boxes in the second frame of the picture. The first step of the fast detection strategy is to set all the pixel values in the E area to 0, set the pixel values in the B and D areas to 0 according to the parity of the number of detections, and mark the area as F. $F = (\text{frame number} \% 2) * B + ((\text{frame number} + 1) \% 2) * D$. The new detection area is marked as G. Then $G = A - E - F$. Through the quick detection, the second step is to filter out the processed picture through MTCNN to detect whether there is a new face. As is shown in the formula of G, the more faces, the larger the E area, the smaller the G area, and the less time it takes to detect. When generating candidate boxes, F regions and E regions do not generate candidate boxes, which greatly reduce the calculation amount of the face candidate boxes detected by the PNet network and avoids the repeated detection of the same target. Each face only needs to be detected once in a multi-target scene. It only takes a small amount of time to achieve the multi-target detection and tracking effect.

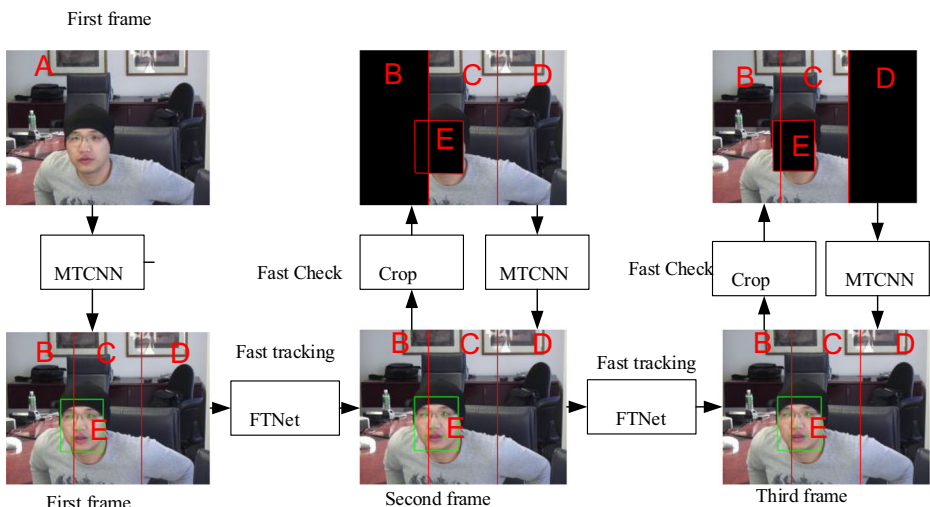


Fig. 4 Flow chart of fast Detection

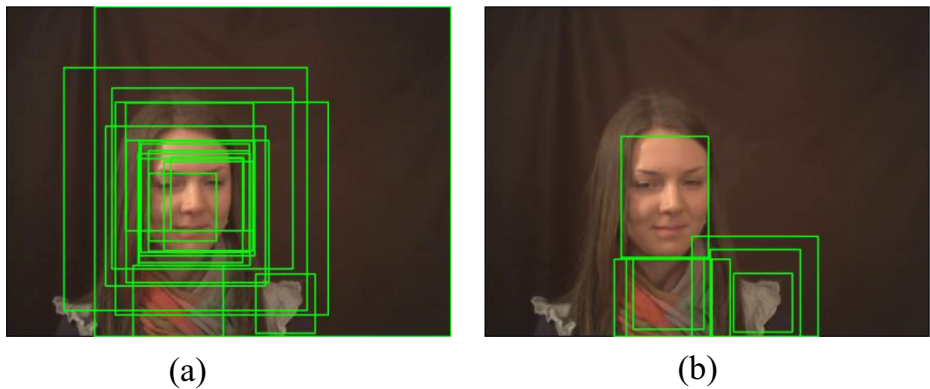


Fig. 5 Comparison of candidate box generation in fast detection

In Eq. 5, matrix A is the feature matrix detected by PNet in the original image, and matrix B is the feature matrix identified by PNet in the image processed by the rapid detection strategy. Matrix B has more 0 than A, As shown in the Fig. 5. picture (a) is the result of the candidate boxes generated in the original image after NMS processing, and the threshold is 0.7. picture (b) is the result of the candidate frame created after using the fast detection processing after NMS processing, and the limitation is 0.7. It is obvious that the rapid detection strategy dramatically reduces the generation of invalid candidate frames.

$$A = \begin{matrix} & x_{11} & x_{12} & x_{13} \cdots & x_{1j} \\ x_{21} & x_{21} & x_{22} & x_{23} \cdots & x_{2j} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_{i1} & x_{i1} & x_{i2} & x_{i3} \cdots & x_{ij} \end{matrix} \gg B = \begin{matrix} & x_{11} & 0 & x_{13} \cdots & 0 \\ x_{21} & x_{21} & x_{22} & 0 \cdots & x_{2j} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_{i1} & 0 & x_{i3} \cdots & x_{ij} \end{matrix} \quad (5)$$

3.3 Fast face recognition

This paper chooses MobileFaceNet and MMFNet for comparison. As shown in Table 2, the reduced network has a lower accuracy than MobileFaceNet, but the model size is reduced by 45%, and the feature extraction speed is reduced by 25%. Different from the literature 14, this paper directly uses the features extracted by MMFNet for feature matching. After the target facial features are extracted, the similarity between the target facial features and all the features in the element database is calculated. In this paper, the cosine similarity is used to represent the similarity of features. The greater the cosine similarity, the higher the feature similarity. Find the most consistent feature in the feature database based on the similarity and determine the target information.

Table 2 Comparison of MobileFaceNet and MMFNe

Net	LFW/%	CFP-FP/%	AgeDB-30/%	Model size/MB	Time/ms
MobileFaceNet	99.20	88.8	91.28	4.0	24
MMFNet	99.12	87.92	90.42	2.2	18

3.4 Algorithm steps

The overall steps of the algorithm are as follows:

Initialize the face collection Facebox, Newface, Checkedface

For every frame in video as *img* DO

if Facebox==0

Facebox $\leftarrow$ MTCNN face detection

else

Facebox $\leftarrow$ Facebox corrected by FTNet, The correction amount is recorded as *R*

for *i* in Size(Facebox)

if $R_i < 50$

Clear Checkedface

add Facebox_{*i*} to Checkedface

img1 $\leftarrow$ fast detection Step A in *img*

Newface $\leftarrow$ MTCNN face detection in img1

Facebox $\leftarrow$ Facebox+Newface

for face in Facebox

if face in Checkedface

Face Recognition in face

else

continue

4 Experiment

4.1 Experimental environment and data set

The FTNet model training server is a 64-bit CentOS system of version 7.5. The CPU is Intel Xeon E5 2620 v4 with four Tesla V100 graphics cards of 16GB, and the memory size is 64GB.

The training data set of FTNet is WilderFace.[27] The dataset contains 32,203 images and 393,703 face tags. 40% of the data is the train set, 10% of the information is the validation set,

Table 3 Training verification data set of FTNet

	Positive	Negative	Part
Training Set	156,728/189530	470,184/975229	156,728/547211
Validation Set	10,000	10,000	10,000

Table 4 Test data set of OTB

Sequence	Framesize	Sequence characteristics	Total frames
David	320*240	IV, SV, OCC, DEF, MB, IPR, OPR	770

and 50% of the data is the test set. The specific settings of the FTNet training data set shown in Table 3.

The tracked test environment is a 64-bit Ubuntu system of version 18.04, the CPU is an Intel (R) Core (TM) i5-8400 CPU @ 2.80GHz, and the memory is 2G. Use the NCNN framework to do forward derivation of neural networks. The OTB test set and 300-VW used to evaluate the tracking accuracy, the Xiph data set is used to assess the detection speed, and the camera data stream is used to determine the impact of the face area on the tracking and detection speed.

OTB is an evaluation standard for the test target tracking algorithm proposed by Wu Yi. The OTB data set includes a sequence of 50 frames and a sequence of 100 frames, which are called OTB50 and OTB100. Different data sets also marked with different attributes, which can represent common difficulties in the field of target tracking. For example, IV (Illumination Variation) indicates changes in lighting, SV (Scale Variation) indicates changes in scale, OCC (Occlusion) indicates occlusion, DEF (Deformation) indicates deformation, MB (Motion Blur) indicates motion blur, IPR (In-Plane Rotation) represents in-plane rotation and OPR (Out-of-Plane Rotation) represents out-of-plane rotation. The following face video from OTB for tracking evaluation shown in Table 4. David's test set includes various scenarios such as IV, SV, OOC, DEF, MB, IPR, OPR and so on. Used to evaluate tracking accuracy will be closer to the complex reality.

The Wild (300-VW)[24] dataset includes 50 training videos and 64 test videos. The test videos divided into three categories, which represent facial videos in three different scenes. As shown in Table 5, five videos are taken from each of the three groups to evaluate the tracking effect.

The Xiph data set is a commonly used test set for target tracking. In this paper, three videos with multiple targets selected from Xiph as the test set to test the speed of detection and monitoring. As shown in Table 6, the frame size is 1280 * 720, and the test set contains 1, 3, and 4 human faces, respectively.

Gradient descent method of the FTNet is Adam, momentum is 0.9, and momentum2 is 0.999, delta is 1e-8, base_lr is 0.01, and weight_decay is 0.0005. MTCNN parameters set the minimum detection face parameter to 100, and the face thresholds of the three cascaded network PRO networks in the fast detection strategy are 0.6, 0.7, and 0.8, respectively.

Table 5 Data set of 300-VW

type	Video ID				
One	1	7	13	19	25
Two	32	36	40	44	48
Three	51	54	57	60	63

Table 6 Data set of Xiph

Video Name	Face Num	Frame Size
Vidyo1_720p_60fps.y4m	3	1280*720
Vidyo4_720p_60fps.y4m	1	1280*720
FourPeople_1280*720p_60.y4m	4	1280*720

4.2 Experimental analysis

The tracking effect of the face in the Min OTB test set measured by the accuracy map and the success rate map. However, the tracking target of the David test set is the human head, and the tracking target of the algorithm in this article is the human face, so that the tracking effect can be measured by the success rate. As shown in Fig. 6, when the threshold is 0.7, the success rate reaches 92.2%. Compared with CXT,[6] VTS,[14] LSK,[17] VTD,[13] MIL,[1] OAB,[8] BSBT,[22] KMS,[4] Struck,[9] SiamRpn[15], the FTNet performs better in complex environments.

The IOU index measures the accuracy of 300-VW face tracking. IOU is the overlap ratio of the target region generated by the model and the original labeled part. Experiments show that if IOU is greater than 0.7, the face tracking result is correct; otherwise, the face tracking fails. On the wild test set, we compared three tracking algorithms, camshaft, particle filter, and FTNet tracking strategy. As shown in Table 7, the FTNet tracking results reached 99.4%, 99.6%, and 100%, respectively. In the test set of the 300-VW, the motion of the target is small, and there is only one target. The accuracy of FTNet tracking is close to 100%. In all three classifications, the accuracy of FTNet is much higher than the accuracy of camshaft and particle filter algorithms.

The test performed on the Xiph dataset, and there were 1, 3, and 4 human face targets in the test videos. The test results shown in Table 8. Through the sparse matrix processed by the fast detection strategy, the number of candidate frames generated is greatly reduced, which makes

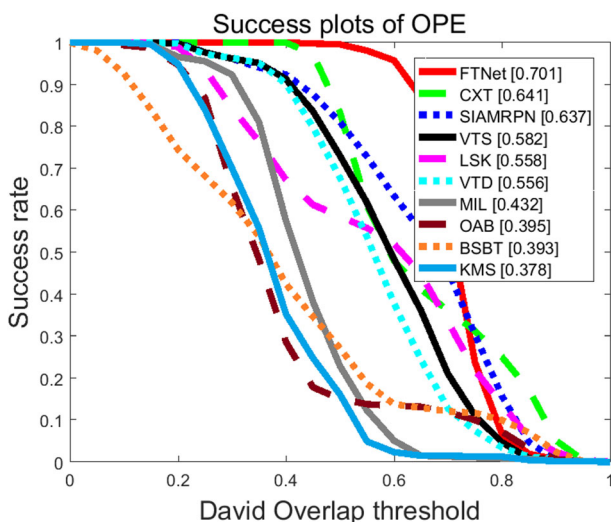
**Fig. 6** Tracking accuracy evaluation Of FTNet

Table 7 Success rate results on videos of min 300-VW

Video id	camshift	particle filter	FTNet
1	72	85	100
7	48	68	100
13	2	56	98
19	57	85	100
25	100	100	99
Average	76.2	87.87	99.4
32	1	60	100
36	100	100	100
40	91	95.6	100
44	8	48	98
48	97	100	100
Average	78.8	89.86	99.6
51	42	78	100
54	0	63	100
57	1	40	100
60	33	56	100
63	2	86	100
Average	52.71	79.14	100

MTCNN reduce the time spent in the detection phase by 18.04 ms, 22.2 ms, and 32.36 ms, respectively. Fast tracking through FTNet takes 1.06 ms, 1.2 ms, and 2.09 ms, respectively. The three test sets for rapid detection and MTCNN reduce overall time consumption by 18.04 ms, 21 ms, and 30.27 ms, respectively. In multi-target tracking and detection scenarios, the number of targets increases and the time consumption through the FTNet phase increases. But because of the rapid detection strategy, the detection image matrix will become more sparse as the number of targets increases. So the time required for the early detection phase will decrease. When the amount of the goal is increased, the overall speed will not decline because of this, and the total time will reach a precise balance.

The tracking results of FTNet on multi-target video sequence shown in Fig. 7. These frames exhibit pose variations, scale changes, and varied backgrounds. As demonstrated by experiments conducted, the FTNet is good at in face tracking tasks, achieving the best performance among the evaluated methods.

During target tracking, the area of the face area is inversely proportional to the area of face detection. The area of the face area will greatly affect the tracking speed. Table 9 shows the evaluation results about the impact of face size on tracking speed in the

Table 8 Fast detection time-consuming comparison

Video name	Frame size		time/ms
Vidyo4_720p_60fps.y4m	640*480	MTCNN	25.66
		Fast detection + MTCNN	7.62
		Fast tracking FTNet	1.06
Vidyo1_720p_60fps.y4m	640*480	MTCNN	32.31
		Fast detection + MTCNN	10.11
		Fast tracking FTNet	1.20
FourPeople_1280*720p_60.y4m	640*480	MTCNN	39.73
		Fast detection + MTCNN	7.37
		Fast tracking FTNet	2.09

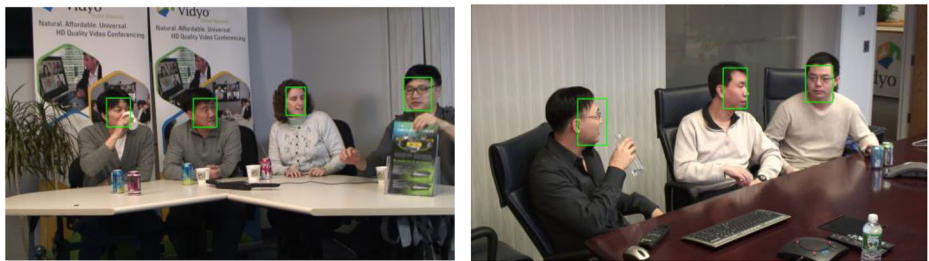


Fig. 7 Multi-target face tracking result

camera data stream. The face sizes are 340×260 and 170×130 respectively. Speed is the average of 200 frames of tracking speed. It can be seen that when the face area is reduced by $1/4$, the tracking speed increased nearly twice. As the detection area becomes more extensive, the detection speed decreases accordingly. But the overall speed is still fast.

5 Conclusion

This paper proposes a multi-target real-time detection, tracking, and recognition algorithm, which includes three strategies: fast detection, fast-tracking, and quick recognition. First, the human faces are detected by a rapid detection strategy, and MMFNet is used to extract features quickly. Finally, fast face tracking through FTNet. The algorithm avoids the repeated detection and recognition of the same target, and the tracking is stable. It successfully converts the multi-target scene into a single target scene. FTNet achieved high tracking accuracy on OTB and 300_VW, and the tracking speed reached 102fps on the Xiph and camera data test set, which improved the tracking speed while ensuring high tracking accuracy. Although MMFNet has reduced efficiency in face recognition, the model size reduced by 45%, and the recognition speed has increased by 25%. The experimental results indicate that the strategy proposed in this paper balances out the problem of time-consuming multi-target tracking and detection in multi-target scenarios. It has sufficient accuracy and real-time performance in detecting and identifying human faces in multi-target scenes.

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Table 9 Target area impact evaluation

Target size	Tracking speed/FPS	detection speed/FPS
340*260	312	152
170*130	666	125

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